

Spotify ⇄ YouTube Music Sync

Comprehensive Installation & Operator Manual

Version 2.0 • Web-Based Zero-CLI Setup • Bi-Directional Transfer & Matching Engine

Who is this guide for? Anyone who wants to transfer their music libraries seamlessly between Spotify and YouTube Music. You **do not** need programming experience or terminal skills. Everything is fully guided with 1-click launchers and an intuitive web interface.

1. Overview & Capabilities

Spotify ⇄ YouTube Music Sync is a high-accuracy, 100% private, self-hosted music bridge. Unlike online services that charge recurring subscription fees, limit transfers to 500 tracks, or resell user data, this application runs entirely locally on your computer.

Spotify → YouTube Music

Exports Spotify playlists and reconstructs them in your YouTube Music library with 99%+ accuracy, automatically weeding out bootlegs, live knockoffs, and karaoke tracks.

YouTube Music → Spotify

Scans YouTube Music playlists, resolves exact canonical tracks on Spotify, and exports direct URI files or creates Spotify playlists with 1-click.

100% Free & Private

No track limits, no monthly fees, and no remote servers. All your tokens, credentials, and track databases remain stored exclusively on your local hard drive.

2. System Prerequisites

Before launching the tool, ensure your system meets the following standard requirements:

1. **Operating System:** Windows 10/11, macOS (Intel/Apple Silicon), or any modern Linux distribution.
2. **Python 3.10 or higher:** Download from python.org/downloads if not installed.

Important for Windows Users: During Python setup, make sure to check the box "Add Python to PATH" before proceeding.

3. **Web Browser:** Google Chrome, Microsoft Edge, Mozilla Firefox, Brave, or Safari.
4. **Google Account:** A standard free Google account associated with your YouTube / YouTube Music profile.

3. Getting Started: Download & 1-Click Launch

Step 1: Download & Extract

1. **Download the Project ZIP:**
<https://github.com/chaos-152/spotify-ytmusic-sync/archive/refs/heads/main.zip>
2. Find the downloaded file `spotify-ytmusic-sync-main.zip` in your Downloads folder.
3. Right-click it and choose **Extract All** (or double-click on macOS).
4. Open the extracted folder. You will see files like `run.bat`, `run.sh`, and folders named `backend` and `frontend`.

Step 2: Start the Tool (1-Click Automated Launchers)

You never need to open a command line manually or configure virtual environments by hand:

Platform	Action	What Happens
Windows	Double-click <code>run.bat</code>	Installs dependencies silently, initializes database, starts the server, and automatically pops open your browser.
macOS / Linux	Run <code>bash run.sh</code> or double-click <code>run.sh</code>	Configures virtualenv, installs requirements, launches server on port 8000, and opens your default browser.

Local Dashboard: Opens automatically at `http://127.0.0.1:8000`. Leave the launcher terminal window open while using the app.

4. First-Run Setup: Web-Based Credentials (OAuth Visual Guide)

The setup flow is **100% web-based**. When launching for the first time, a setup modal will automatically guide you step-by-step through configuring your free developer keys.

Step 1 of 2: YouTube Music (Google Cloud OAuth)

To let the app manage playlists on your YouTube Music account without quotas or subscription fees, Google provides a free personal API key:

1. Open [Google Cloud Console](#) and sign in.
2. Click **Select a Project** > **New Project**, enter `Playlist Sync`, and click **Create**.
3. In the top search bar, search for **YouTube Data API v3**, click it, and click **Enable**.

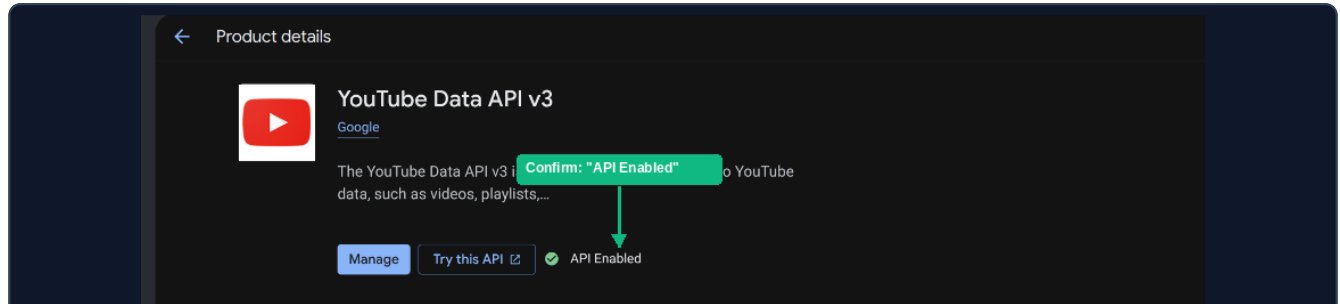


Figure 1: Confirm that the **YouTube Data API v3** displays "API Enabled" in your project.

1. Configure the **OAuth Consent Screen** (from the left menu):
 - User Type: Choose **External** > click **Create**.
 - Enter an App name (e.g. `Playlist Sync`) and your user support email > click **Save and Continue**.
 - Under **Test users**, click **+ Add Users**, enter your own Gmail address, and click **Save**.

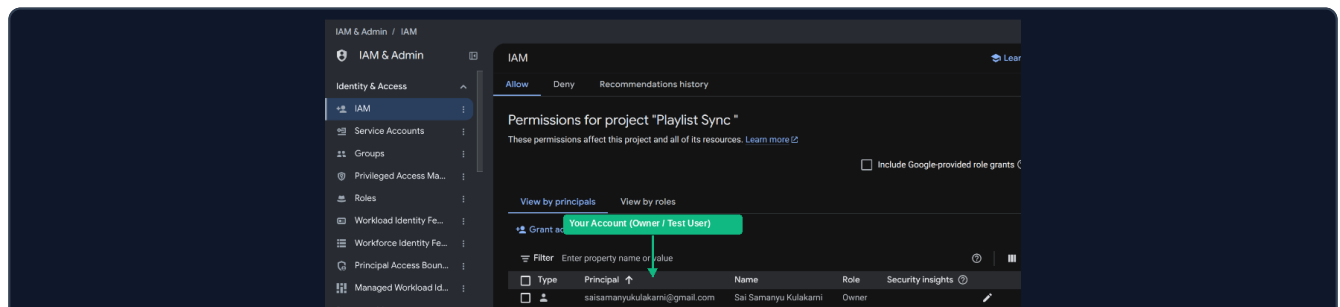


Figure 2: Ensure your project is named **Playlist Sync** and your personal account is confirmed as an authorized user.

1. Generate API Credentials:
 - On the left menu, click **Credentials** > **+ Create Credentials** > **OAuth client ID**.
 - For *Application type*, choose **TVs and Limited Input devices** (*Crucial: do not pick Web application*).
 - Click **Create**. Google generates your **Client ID** and **Client Secret**.

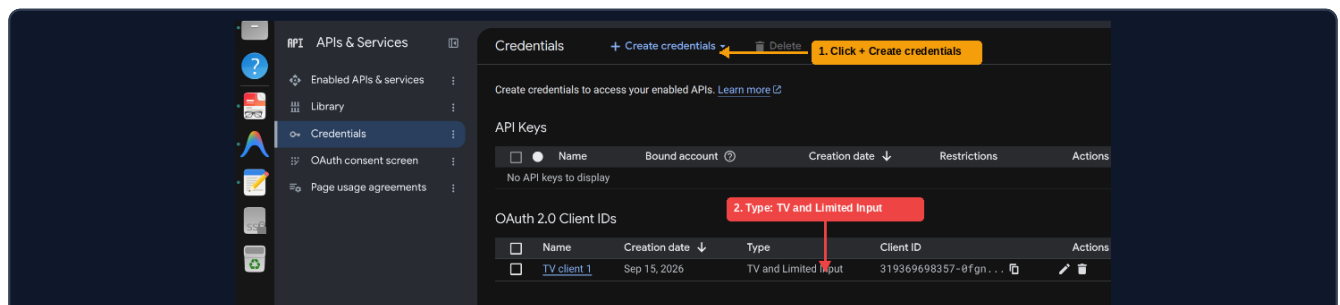


Figure 3: Click **+ Create credentials** > **OAuth client ID**, and verify type is **TV and Limited Input**.

1. Paste the Client ID and Client Secret into the web dialog in your browser and click **Save & Configure**.
2. Click **Connect YT Music**:
 - Click the displayed link to `https://www.google.com/device` and enter the 8-character verification code shown on your screen.

- *Google Unverified App Screen*: Click **Advanced** > **Go to Playlist Sync (unsafe)** > **Allow**. This warning appears simply because your private personal key is in testing mode.
- Return to the app tab — the status indicator will turn green: **Connected**.

Step 2 of 2: Spotify Developer Credentials (Optional / Reverse Sync)

Directly after completing Step 1, the modal moves to **Step 2 of 2 (Spotify Setup)**:

1. Navigate to developer.spotify.com/dashboard and log in.
2. Click **Create App**. Enter an app name and set the **Redirect URI** to:
`http://127.0.0.1:8000/api/spotify/callback`
 (⚠ *Note: Spotify requires the numeric literal `127.0.0.1`; `localhost` is rejected*).
3. Copy your Spotify **Client ID** and **Client Secret**, paste them into the modal, and click **Finish Setup & Start Syncing**.

Dismiss & Resume Point: If you dismiss the setup modal mid-way (e.g., after Google is configured), the app automatically remembers your progress and re-routes directly to Step 2 on reload.

5. How to Sync: Spotify → YouTube Music (Forward Sync)

Transfer any Spotify playlist to YouTube Music in 3 effortless steps:

1. **Export Playlist from Spotify:**
 - Visit exportify.net in your browser and sign in with Spotify.
 - Click **Export** next to the playlist you want. A clean `.csv` file will download immediately.
2. **Import into Sync Engine:**
 - Return to the app tab (`http://127.0.0.1:8000`).
 - Drag and drop the exported CSV file into Section 2, or click to browse.
 - The playlist title populates automatically. Click **Import Playlist**.
3. **Preview & Execute Sync:**
 - Under **Section 3 (Imported Playlists)**, click **Pre-Sync Preview** to view matched YouTube tracks, duration deltas, and similarity scores.
 - Notice the subtle `? similar track nearby` badge if multiple versions exist in the playlist.
 - Click **Sync Now**. A real-time progress bar tracks progress.
 - When complete, click **Open in YT Music** ↗ to listen immediately!

6. How to Sync: YouTube Music → Spotify (Reverse Sync)

Migrating playlists back to Spotify is supported through either direct 1-click sync or universal CSV export:

1. In the navigation bar, switch to the **Move to Spotify** tab.
2. Select any YouTube Music playlist from your library or enter a playlist link.
3. Click **Fetch Tracks & Match**. The engine queries Spotify's catalog to find the exact matching master releases.
4. **Choose your import method:**
 - **Method A (Direct 1-Click Sync):** Click **Sync Directly to Spotify**. A new playlist is automatically generated and populated in your Spotify library.
 - **Method B (Desktop Quick Paste):** Click **Download CSV**. Open the file, copy the column of Spotify URIs (e.g. `spotify:track:...`), open Spotify Desktop, click an empty playlist, and press `Ctrl+V` (Windows) or `Cmd+V` (macOS).
 - **Method C (Web Importer - Spotlitr):** Upload the generated CSV to spotlitr.com to import tracks via web browser.

7. Troubleshooting & FAQ

Issue	Cause & Quick Resolution
"Session expired" after 7 days	Because personal Google Cloud projects default to "Testing" mode, OAuth refresh tokens expire after 7 days. Simply click Connect YT Music and re-enter the 8-character code at <code>google.com/device</code> . All playlists remain intact.
Browser doesn't open automatically	On some systems, background browser popups are blocked by OS permissions. Open Chrome, Edge, or Firefox manually and navigate to <code>http://127.0.0.1:8000</code> .
Spotify 403 Forbidden Error	Spotify Developer apps operate in "Development Mode" by default. In your Spotify Developer Dashboard under User Management , ensure you add the exact email address of your personal Spotify account.
Accidental Duplicates Detected	The engine includes an 11-factor duplicate detection filter. Tracks sharing artist and normalized title within a 15-second duration tolerance are automatically identified to prevent duplicates.
Data Privacy & Local Storage	All tokens, secrets, cached links, and playlist histories are saved in <code>backend/app.db</code> on your local disk. No third-party servers ever receive your listening history or tokens.

Stopping the Server: When finished using the tool, simply close the terminal launcher window. Your configuration and playlists are preserved for your next session.